12.24 The Central Limit Theorem

WORKSHEET

The question

STEP III 2008, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The density $\frac{1}{2}e^{-|x|}$ is symmetric. What does it look like? What is its mean?

STAGE 1 The moment generating function

Covers: the sketch, and the moment generating function

Ask yourself Split the integral for $\mathrm{E}(e^{\theta X})$ at 0. Why do you need $|\theta| < 1$?

Ask yourself Expand $\frac{1}{1-\theta^2}$ as a series to read off the mean and variance.

STAGE 2 The frog's displacement

Covers: the moment generating function of $Y/\sqrt{2n}$

Pause and try The jumps are independent, so what is the moment generating function of their sum Y? What does dividing Y by $\sqrt{2n}$ do to θ?

STAGE 3 The limit

Covers: the limit

Ask yourself Take logarithms and use $\ln(1-u) = -u - \frac{1}{2}u^2 - \cdots$ for small u. Which distribution has the limiting moment generating function?

STAGE 4 The frog's chances

Covers: the least number of jumps

Ask yourself For large n, Y is approximately $\mathrm{N}(0, 2n)$. For a standard normal Z, $\mathrm{P}(|Z| \geq 1.96) = 0.05$. Solve for n.

What's really going on

The general theorem

The same argument works for any distribution with mean 0 and finite variance σ^2 (and a moment generating function, though this can be avoided). Then $\mathrm{M}(\theta) = 1 + \frac{1}{2}\sigma^2\theta^2 + o(\theta^2)$, and

$$\left(\mathrm{M}\left(\frac{\theta}{\sigma\sqrt{n}}\right)\right)^n = \left(1 + \frac{\theta^2}{2n} + o\left(\frac{1}{n}\right)\right)^n \to e^{\theta^2/2}.$$

Only the variance survives the limit: the shape of the original distribution is forgotten. That is why the normal distribution is universal. At university the theorem is proved with the *characteristic function* $\mathrm{E}(e^{i\theta X})$, which always exists, and the convergence of distributions is made precise by Lévy's continuity theorem.

How fast?

The central limit theorem says what happens in the limit, not how quickly. The *Berry–Esseen theorem* bounds the error by a constant times $\frac{\mathrm{E}|X|^3}{\sigma^3\sqrt{n}}$. For a symmetric distribution like the frog's, the approximation is good after a few dozen jumps. For skewed distributions, or in the extreme tails, it can be poor even for large n: the normal approximation underestimates the chance of very rare events, a fact with consequences for insurance and finance.

From de Moivre to Lindeberg

De Moivre found the normal approximation to the binomial in 1733 (Section 12.23), and Laplace extended it to sums of more general variables in 1810, coining the idea that errors of measurement are sums of many small effects. Rigorous proofs came much later: Aleksandr Lyapunov in 1901, and Jarl Lindeberg in 1922, who found

essentially the weakest conditions under which the theorem holds. The name "central limit theorem" was given by George Pólya in 1920, meaning that the theorem is central to probability theory.

Try this next

1. **Exact for two jumps.** Show that $Y = X_1 + X_2$ has density $\frac{1}{4}(1 + |y|)e^{-|y|}$, and check its variance is 4.

2. **Binomial to normal.** For $X \sim \mathrm{B}(1, \frac{1}{2})$ shifted to have mean 0 (taking values $\pm\frac{1}{2}$), find $\mathrm{M}_X(\theta) = \cosh\frac{\theta}{2}$, and show that the moment generating function of a standardised sum of n such variables tends to $e^{\theta^2/2}$.

3. **The frog, exactly.** Use a computer (or the exact density for $n = 2$ as a check) to see how accurate the normal approximation is for $n = 82$.

4. **Why not Cauchy?** The Cauchy distribution (Section 12.22) has no moment generating function. Which step of the argument fails, and what happens to the average of n Cauchy variables?

Further reading

- **Geoffrey Grimmett & Dominic Welsh, *Probability: An Introduction*** (Oxford University Press), the chapter on generating functions and limit theorems.
- **Hans Fischer, *A History of the Central Limit Theorem*** (Springer). From Laplace to Lindeberg.
- **David Spiegelhalter, *The Art of Statistics*** (Pelican), on why the normal distribution is everywhere.

The full guided solution is in Section 12.24:
https://beyondstep.keithmcnulty.org/ch12/clt.html